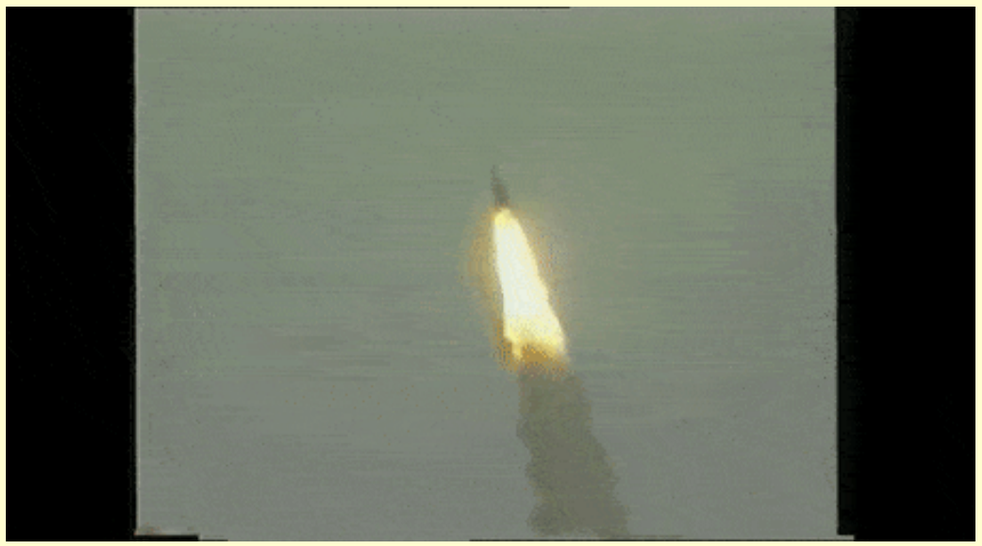
Hardware & Software Verification

John Wickerson

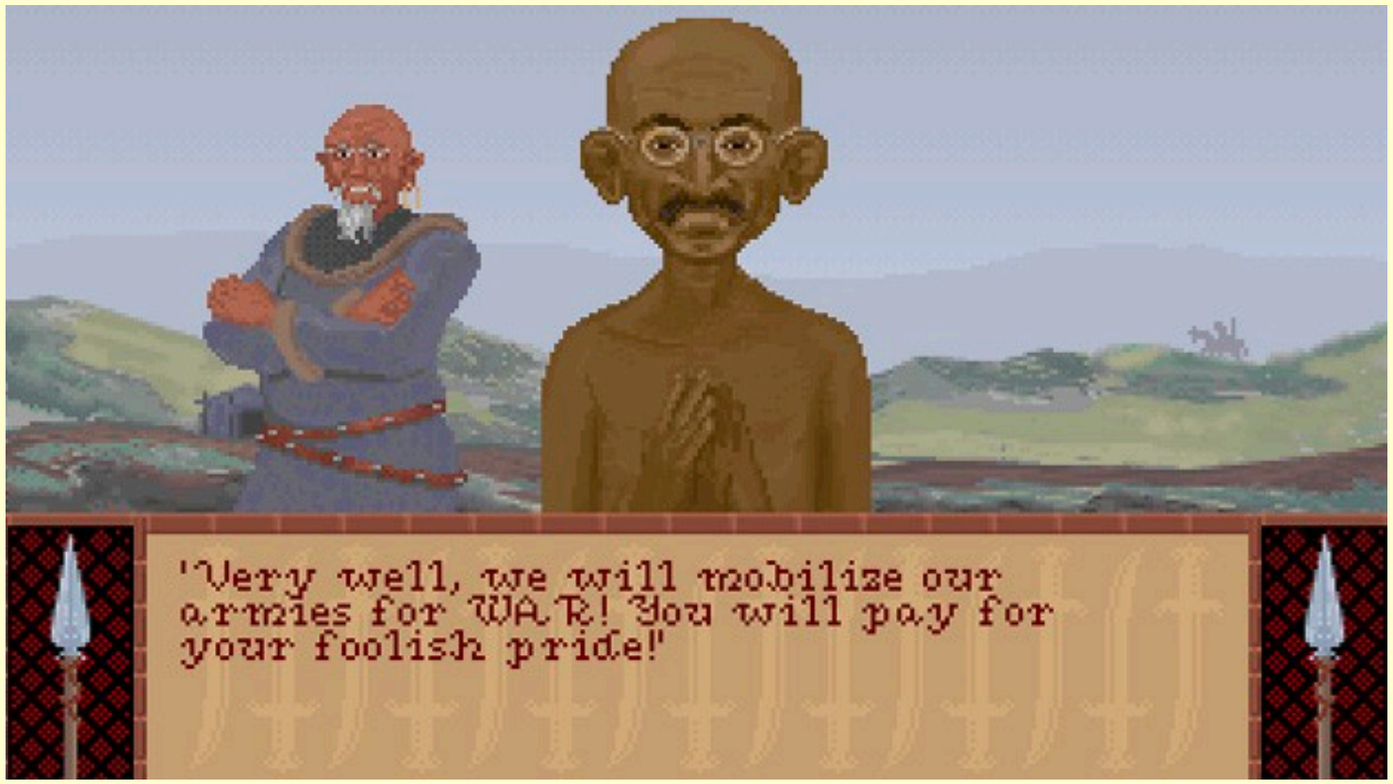
https://github.com/johnwickerson/HSV

Lecture 1
5 October 2026

The correctness problem



```
int main () {
  double a = 4195835;
  double b = 3145727;
  printf ("%f\n", a/b);
}
```





The correctness problem

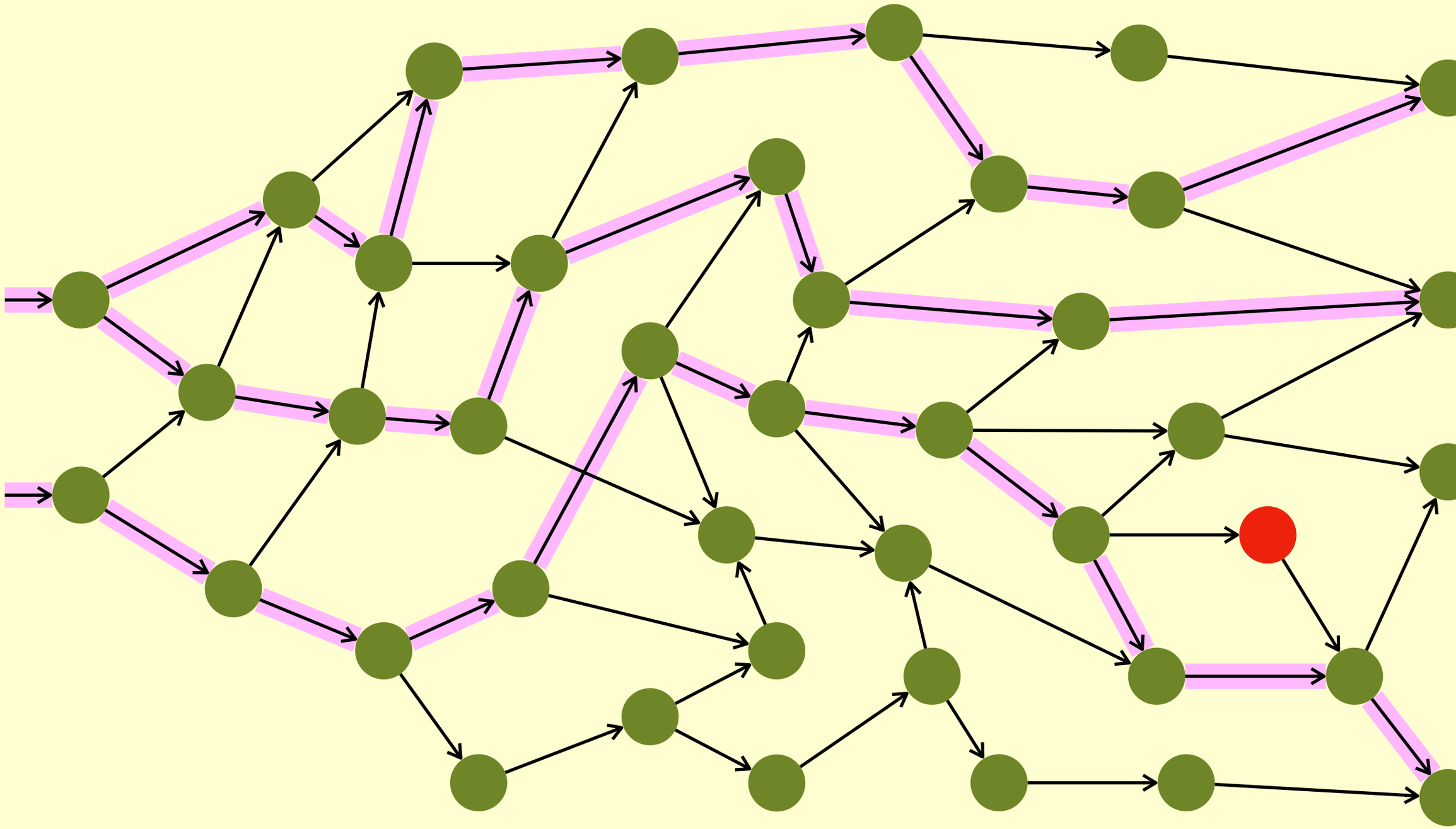
- Computer systems are becoming more **complicated** and more **trusted**.

- This means that being confident that they are correct is increasingly **difficult** and increasingly **important**.

- Traditional **testing** is no longer enough, especially in our manycore era.

- Fortunately there are techniques and tools for **verifying** that a computer system is correct.

The problem with testing



The problem with testing

```c
#include <stdlib.h>
#include <stdio.h>

int main(int argc, char **argv) {
  if (argc > 1 && atoi(argv[1]) == 4242) {
    printf ("KABOOM!\n");
    return 1;
  }
}
```

The problem with testing

```
int main() {
  int x = 0, y = 0;
  int r1 = 0, r2 = 0;

  x = 1;  ||  r1 = y;
  y = 1;  ||  r2 = x;

  if (r1==1 && r2==0) {
    printf ("KABOOM!\n");
    return 1;
  } else {
    printf ("r1=%d, r2=%d\n", r1, r2);
    return 0;
  }
}
```

- Doesn't cope with **nondeterminism** (e.g. from concurrency).
- Testing results may not be **portable**.

Can we do better?

- Rather than testing on many *particular* inputs, construct a general argument for why the program is correct on *all* inputs.



"There is no solution to $a^n + b^n = c^n$ when $n>2$."



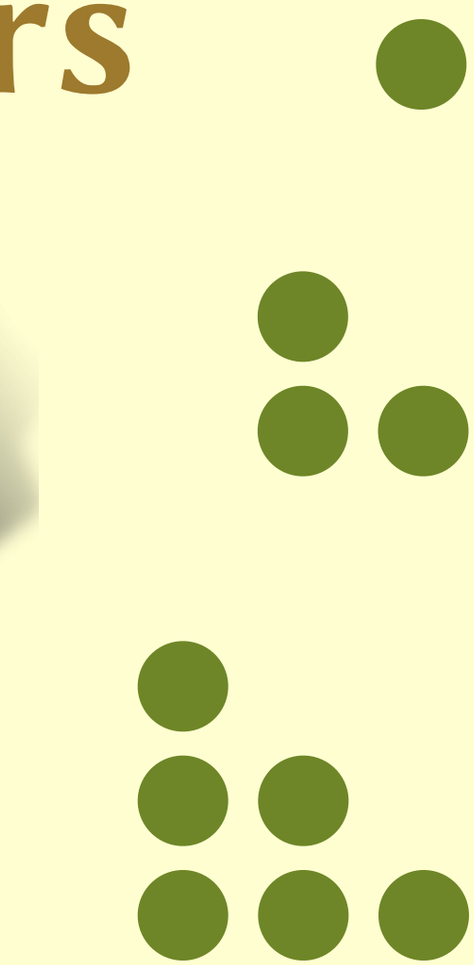
Triangle numbers

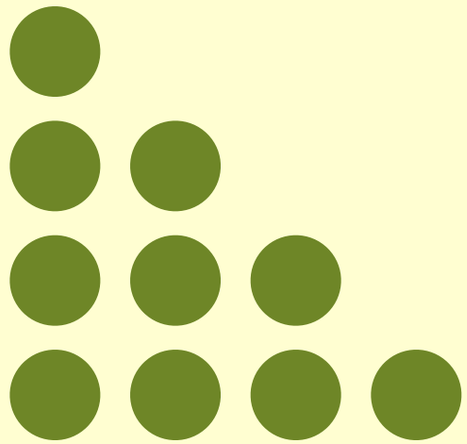
```
#include <stdlib.h>
#include <assert.h>

int triangle(int n) {
  int t = 0, i = 0;
  while (i < n) {
    i = i+1;
    t = t+i;
  }
  assert(t == n * (n+1) / 2);
  return t;
}

int main(int argc, char **argv) {
  int n = atoi(argv[1]);
  printf("%d\n", triangle(n));
}
```







Dafny

- Developed since 2008 by Rustan Leino and others at Microsoft Research, and nowadays mainly at Amazon.

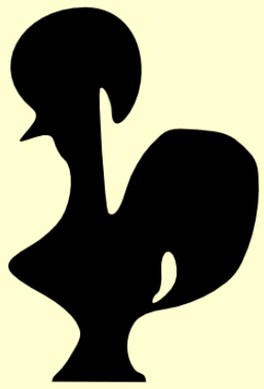




Verifying with Dafny

```
method triangle(n:int) returns (t:int)
  requires 0 <= n
  ensures t == n * (n+1) / 2
{
  t := 0;
  var i := 0;
  while i < n
    invariant t == i * (i+1) / 2
    invariant i <= n
  {
    i := i+1;
    t := t+i;
  }
  return t;
}
```

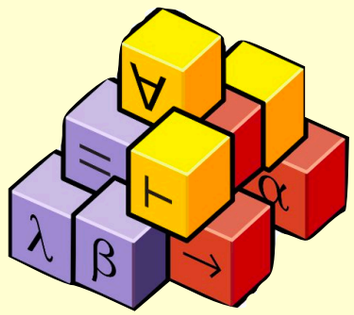
Interactive theorem proving



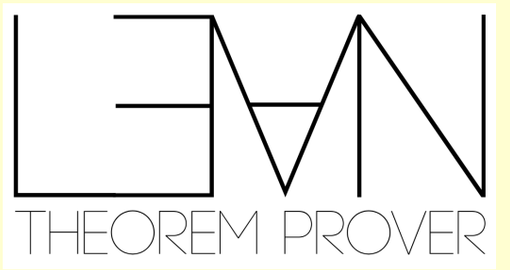
Rocq



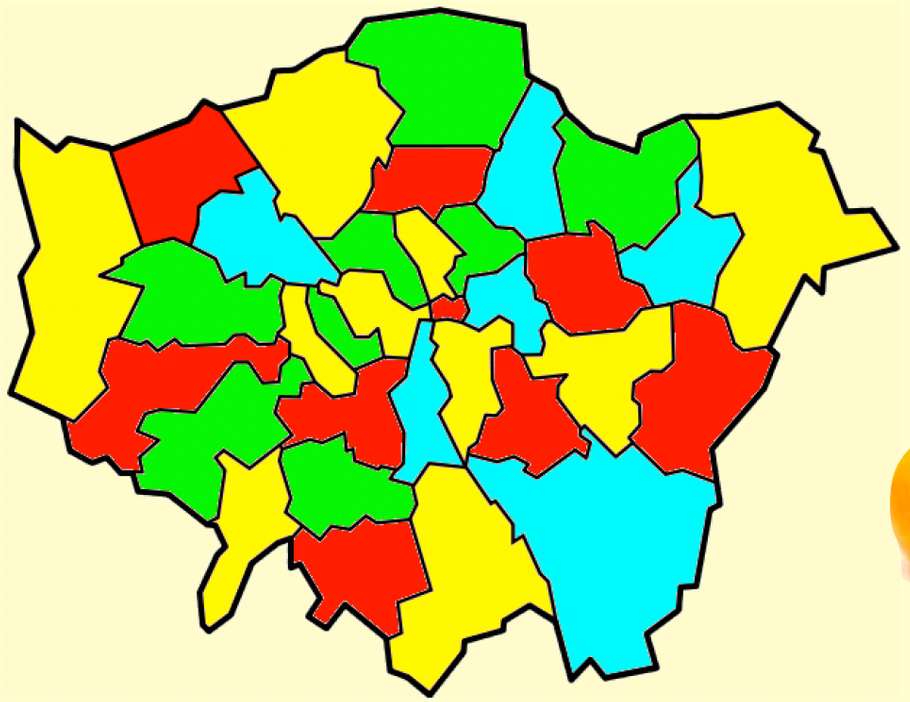
HOL



Isabelle

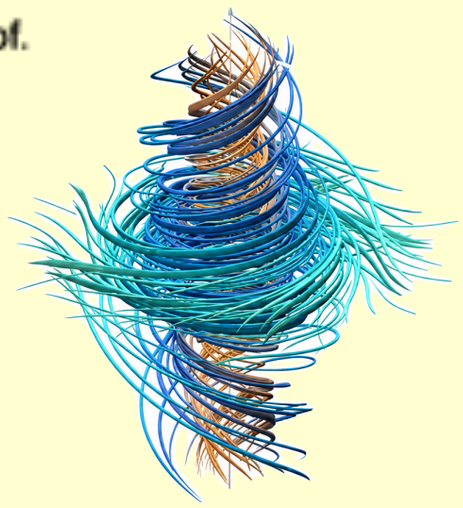


Lean



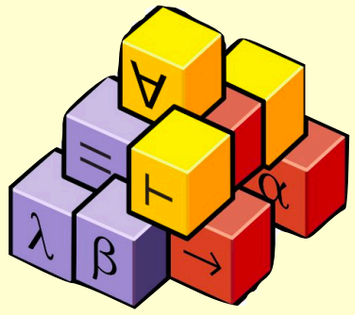




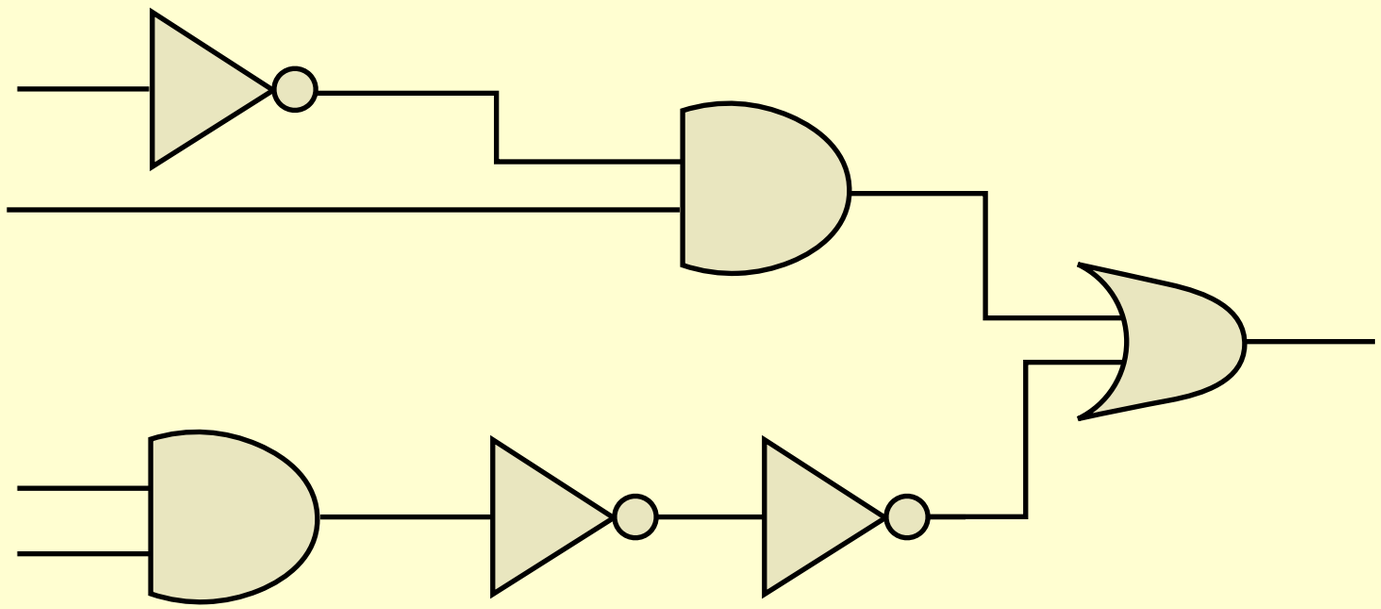


Interactive theorem proving

- *Your task:* prove a correctness theorem about a little logic synthesiser.



Isabelle



Computer-aided proof

Fully manual

Fully automatic

Rocq	**SymbiYosys**	Facebook Infer
HOL	**Dafny**	Astrée
Isabelle	VCC	Model checking

Aims of the module

1. Be able to use Dafny to verify simple programs.

   - **Assessment:** I will give you several (increasingly difficult) programs, and you have to verify them using Dafny.

2. Be able to use Isabelle to conduct simple mathematical proofs.

   - **Assessment:** I will give you several (increasingly difficult) theorems, and you have to prove them using Isabelle.

3. Be able to use SymbiYosys to verify simple Verilog designs.

   - **Assessment:** I will give you several (increasingly difficult) Verilog designs, and you have to verify them using SymbiYosys.

Rough plan

- Introduction
- Proving theorems using Isabelle
- Verifying software using Dafny
- Verifying hardware using SymbiYosys
- Principles of verification
- **Guest lecture from Apple:** date TBC.

Rough plan

w.b.	Monday 4-6pm (room 407)	Thursdays 12-2pm (room 508B)
5 Oct	Introduction	Isabelle 1
12 Oct	Isabelle 1	*Lab*
19 Oct	Isabelle 2	*Lab*
26 Oct	Dafny	Guest lecture from **Apple**
2 Nov	Formal HW verification	SAT / SMT
9 Nov	**Mock test (room 303)**	Temporal logic and BDDs
16 Nov	Hoare Logic	*Lab*
23 Nov	**Test: Isabelle (room 303)**	*Lab*
30 Nov	**Test: Dafny (room 303)**	*Lab*
7 Dec	**Test: Symbiyosys (room 303)**	UVM

Teaching support



Quentin Corradi



Michalis Pardalos

A first proof

- **Theorem.** $\sqrt{2}$ is irrational.

A first proof

- **Theorem.** $\sqrt{2}$ is irrational.

- **Proof.** We proceed by contradiction. Suppose $\sqrt{2}$ is rational. Then we can obtain integers m and n such that $\sqrt{2} = m/n$, where $n>0$ and $\gcd(m, n) = 1$. Squaring both sides gives us $2 = m^2/n^2$, and hence $2n^2 = m^2$. This means that m^2 is even, and hence so is m. Hence, we can obtain an integer m' such that $m = 2m'$. It follows that $2n^2 = (2m')^2$, and hence $n^2 = 2m'$. This means that n^2 is even, and hence so is n. But if both m and n are even, their gcd is at least 2, which contradicts the fact that $\gcd(m, n) = 1$. Thus our original supposition must have been wrong, which means that $\sqrt{2}$ is *not* rational.


DEMO

On AI

- AI is having huge impact on formal verification
- AI can generate Isabelle proofs, and the proof checker means we don't have to worry about hallucinations!
- There is suddenly a huge **demand** for proofs *and* a huge **supply** of them!
- We still need humans to check the *definitions*.
- Feel free to use AI to help you learn, but no AI in the tests!